

Gabriel Cassoli Capellini

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Nationality Brazilian
Webpage Personal Website
Research Interests Gamma-ray Astronomy; Multi-Messenger Astronomy; Astroparticle Physics; High-Energy Astrophysics.

Education

2025-present Master of Science, Department of Astronomy - University of São Paulo (IAG-USP), Brazil

Advisor Prof. Dr. Maria Victoria del Valle

Thesis:

Multiwavelength Modeling of Gamma-ray Emitters: Unveiling the Nature of Unidentified Sources

2020-2024 Bachelor of Science, Astronomy - University of São Paulo (IAG-USP), Brazil

Advisor Prof. Dr. Maria Victoria del Valle

Thesis:

T Tauri stars as Transient Gamma-ray Sources

Technical & Computational Skills

Programming Languages: Python, Fortran, R, OpenMP, IRAF

Tools & Frameworks: LaTeX, Git/GitHub.

Publications & Scientific Output

Submitted Manuscripts & In Preparation

- 2026** del Valle, M.V. and Capellini, G.C. : *Hadronic Transient Gamma-ray Emission from a Protostar*
Submitted to *Astronomy & Astrophysics* (A&A).
- 2026** Capellini, G.C., del Valle, M.V., et al. : *Modeling the non-thermal emission from the protoplanetary nebula IRAS 18443-0231*
In preparation for *Astronomy & Astrophysics* (A&A).
- 2026** L. Supan, et al. : *Cornering the elusive engine powering the high-energy source 4FGL J1846.9-0227*
In preparation for *Nature Astronomy*.

Conference Proceedings

- 2026** Capellini, G.C. and del Valle, M.V. : *Non-thermal leptonic emission from a protoplanetary nebula*
Proceedings of Science (HEPRO IX).
- 2024** del Valle, M.V. and Capellini, G.C. : *On the nature of the transient gamma-ray source associated with a protostar*
Accepted for publication in *Memorie della Società Astronomica Italiana* (MemSAIt).

Local Publications & Academic Theses

- 2026** Capellini, G.C. and del Valle, M.V. : *Are T Tauri stars gamma-ray emitters? Insights from a non-thermal model*
Boletim da Sociedade Astronômica Brasileira, vol. 37.
- 2025** Capellini, G.C. and del Valle, M.V. : *T Tauri stars as transient gamma-ray sources*
Boletim da Sociedade Astronômica Brasileira, vol. 36.
- 2024** Capellini, G.C. and del Valle, M.V. : *Estrelas T Tauri como fontes transientes de raios gama*
Bachelor's Thesis (TCC), Instituto de Astronomia, Geofísica e Ciências Atmosféricas (IAG-USP).

Conferences, Workshops & Presentations

International Conferences & Schools

- Jun. 2026** ICTP-SAIFR School on Astroparticle and Multi-messenger Astrophysics, São Paulo, Brazil.
Poster: *Multiwavelength analysis of a protoplanetary nebula: particle acceleration from radio to gamma-rays.*
- Aug. 2025** High Energy Phenomena in Relativistic Outflows IX (HEPRO IX), Rio de Janeiro, Brazil.
Poster: *A new gamma-ray source? The role of jet-outflow interaction in a protoplanetary nebula.*
- Aug. 2025** CLAF/ICTP-SAIFR Latin-American Astroparticle Physics School, São Paulo, Brazil.
Poster: *The role of a jet-outflow system in gamma-ray emission from a protoplanetary nebula.*
- Jun. 2024** MPIK-CDY School on the Future of Gamma-Ray Astronomy, Heidelberg, Germany.
Poster: *Particle Acceleration in Galactic Sources.*

National & Regional Meetings

- Oct. 2025** XLVIII Annual Meeting of the Brazilian Astronomical Society (SAB), Caxambu, Brazil.
Poster: *Are T Tauri stars gamma-ray emitters? Insights from a non-thermal model.*
- Sep. 2024** XLVII Annual Meeting of the Brazilian Astronomical Society (SAB), Águas de Lindóia, Brazil.
Poster: *T Tauri stars as transient gamma-ray sources.*
- Oct. 2023** XLVI Annual Meeting of the Brazilian Astronomical Society (SAB), Rio de Janeiro, Brazil.
Poster: *Study on Particle Acceleration in Shock Waves in Galactic Sources.*

Institutional Seminars & Academic Symposia

- Sep. 2026** *XI IAG Science Day*, IAG-USP, São Paulo, Brazil.
Talk: *Planetary Nebulae: A New Candidate Gamma-Ray Source Class.*
- Oct. 2025** *Astronomia ao Meio-Dia Seminar*, IAG-USP, São Paulo, Brazil.
Invited Talk: *Galactic Accelerators: the Search for Unidentified Sources.*
- Sep. 2025** *X IAG Science Day*, IAG-USP, São Paulo, Brazil.
Talk: *Jets in Planetary Nebulae: a New Window to Gamma-ray Sources.*
- Oct. 2024** *32nd USP International Symposium on Scientific and Technological Initiation (SIICUSP)*, São Paulo, Brazil.
Poster: *T Tauri stars as transient gamma-ray sources.*
Honors: Awarded International Stage Qualification.
- Sep. 2024** *IX IAG Science Day*, IAG-USP, São Paulo, Brazil.
Talk: *T Tauri Stars as Transient Gamma-ray Sources.*
- Nov. 2023** *VIII IAG Science Day*, IAG-USP, São Paulo, Brazil.
Talk: *Particle Acceleration in Galactic Sources.*
- Oct. 2023** *31st USP International Symposium on Scientific and Technological Initiation (SIICUSP)*, São Paulo, Brazil.
Talk & Poster: *Particle Acceleration in Galactic Sources.*
Honors: Awarded 2nd Place (Departmental Stage) & International Stage Qualification.

Research experience

- Mar.2026 - present** University of São Paulo (IAG-USP), São Paulo, Brazil
Member of the ASTRI-Mini Array Data Analysis Working Group
Supported by FAPESP (No.2025/06590-1)
- Oct.2025 - present** University of São Paulo (IAG-USP), São Paulo, Brazil
Development of the Non-thermal Spectrum Calculator (NOTSC) software
Supported by FAPESP (No.2025/06590-1)
- Mar.2025 - present** University of São Paulo (IAG-USP), São Paulo, Brazil
Multiwavelength Modeling of Gamma-ray Emitters: Unveiling the Nature of Unidentified Sources
Supported by CAPES (88887.142048/2025-00) and FAPESP (No.2025/06590-1)
- Feb.2024 - Dec.2024** University of São Paulo (IAG-USP), São Paulo, Brazil
T Tauri Stars as Transient Gamma-ray Sources
Supported by FAPESP (No.2022/16530-8)
- Feb.2023 - Dec.2023** University of São Paulo (IAG-USP), São Paulo, Brazil
Particle Acceleration in Galactic Sources
Supported by FAPESP (No.2022/16530-8)

Teaching experience

- Mar. 2025 - present** University of São Paulo (IAG-USP), Brazil
Participation in the Interstellar Medium and Cosmic Rays Astrophysics (IMCA) Group Discussions - USP

May 2026 University of São Paulo (IAG-USP), Brazil
Invited Lecturer in The Graduate Course 'Radiative Processes' - USP

Aug.2024 - University of São Paulo (IAG-USP), Brazil
Dec.2024 *Teaching Assistant in High Energy Astrophysics - USP*

Feb.2024 - University of São Paulo (IAG-USP), Brazil
Jul.2024 *Teaching Assistant in Introduction to Astronomy I - USP*

Feb.2022 - University of São Paulo (IAG-USP), Brazil
Jul.2022 *Teaching Assistant in Introduction to Astronomy I - USP*

Awards & Honors

2023 **2nd Place Best Undergraduate Research Award**, Department of Astronomy (IAG-USP).
 Awarded 2nd best research presentation at the departmental stage of the 31st USP International Symposium on Scientific and Technological Initiation (SIICUSP) for the project *"Particle Acceleration in Galactic Sources"*.

2023 **International Stage Qualification**, 31st SIICUSP, University of São Paulo.
 Selected to advance to the final international stage of the university-wide research symposium.

2024 **International Stage Qualification**, 32nd SIICUSP, University of São Paulo.
 Selected to advance to the final international stage of the university-wide research symposium.

2024 **Academic Standing – Top 3 Ranking**, Department of Astronomy (IAG-USP).
 Ranked 3rd out of 18 students in the undergraduate class based on overall GPA / academic record.